

Exercise - Week 4: MATH1061/7861

Duration: 30 minutes

2 Questions - 10 marks

Topics: Statements with Multiple Quantifiers, Direct Proofs and Counterexamples, and Proof by Contradiction

Student Name:

Student number: |_|_|_|_|_|_|_|_|

*Your student number should be 8 digits and **not** start with an s. e.g. 41234567*

Problems. Prove or disprove the following statements:

(1) $\forall m \in \mathbb{Z}, \exists n \in \mathbb{Z}$, such that $m + n$ is odd and mn is even. **(3 marks)**

(2) For all $x \in \mathbb{Z}$, there exists $y \in \mathbb{Z}$ such that $x = 3y + 5$. **(3 marks)**

- (3) There is no smallest positive real number. **(4 marks)**